

Multiple Choice

1. **Answer:** C

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: C - True, False

2. **Answer:** B

Options: A) If A is open, then $A = \mathbb{R}$.; B) If A is closed, then $A = \mathbb{R}$.; C) If A is uncountable, then $A = \mathbb{R}$.; D) If A is uncountable, then A is open.

Note: Correct option: B - If A is closed, then $A = \mathbb{R}$.

3. **Answer:** C

Options: A) 4; B) 8; C) 12; D) 6

Note: Correct option: C - 12

4. **Answer:** C

Options: A) -11; B) 0; C) 11; D) $33/2$

Note: Correct option: C - 11

5. **Answer:** C

Options: A) The ring of complex numbers; B) The ring of integers modulo 11; C) The ring of continuous real-valued functions on $[0, 1]$; D) The ring $\{a + b * \sqrt{2} : a \text{ and } b \text{ are rational numbers}\}$

Note: Correct option: C - The ring of continuous real-valued functions on $[0, 1]$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '2'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0, 1, and 2 only'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $\sqrt{34}$ '.

Long Form Response

11. **Answer:** The first ten positive composite integers are 4, 6, 8, 9, 10, 12, 14, 15, 16, 18. The desired quotient is $\frac{4 \cdot 6 \cdot 8 \cdot 9 \cdot 10}{12 \cdot 14 \cdot 15 \cdot 16 \cdot 18}$. After cancellation, we get $\frac{1}{14 \cdot 3} = \frac{1}{42}$.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** f is one-to-one.. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key